RetailPulse Enterprise Intelligence

Comprehensive Technical & Strategic Project Documentation | Zidio Development Internship

PRODUCTION-READY | DOCKERIZED & CLOUD DEPLOYED

CORE ENGINE	UI FRAMEWORK	DEPLOYMENT
Python 3.10	Streamlit	Docker / Cloud

1. Executive Summary

RetailPulse is an advanced, production-grade retail analytics and demand forecasting command center built to solve real-world inventory management and customer retention bottlenecks. By integrating state-of-the-art machine learning models with a high-performance interactive web dashboard, RetailPulse empowers retail leadership to make data-driven decisions regarding stock optimization, promotional targeting, and customer lifetime value.

2. Architecture & Tech Stack

Designed with modularity and scalability in mind, the platform uses a clean separation of concerns. The codebase is organized into core dashboard modules, dedicated machine learning pipelines, and containerized deployment artifacts:

- **Frontend UI:** Streamlit reactive multi-page dashboard featuring custom CSS styling and responsive layout containers.
- **Analytics & Modeling:** Pandas, NumPy, Scikit-Learn, and Facebook Prophet for time-series forecasting and customer clustering.
- **Containerization:** Fully isolated Docker container (`retailpulse-app`) orchestrated via Docker Compose on port 8501.

3. Core Analytical Modules & Findings

The application processes transactional datasets through three mission-critical analytical pipelines, outputting persistent reports saved within the system architecture.

Analytical Module	Methodology / Algorithm	Business Artifact Generated
RFM Customer Segmentation	Recency, Frequency, Monetary behavioral clustering	`customer_segments_rfm.png`
Demand Forecasting	Facebook Prophet time-series trend analysis	`prophet_demand_forecast.png`
Data Drift Detection	Statistical distribution comparison across historical batches	`drift_detection_report.csv`

4. Key Insights & Visual Deliverables

The generated visual artifacts verify model convergence and operational stability. RFM clustering successfully isolates high-value VIP cohorts from churning customers, enabling hyper-targeted marketing. Meanwhile, Prophet demand forecasts project inventory requirements with high statistical confidence, preventing both stockouts and overstock overhead.

5. Conclusion & Business Impact

RetailPulse bridges the gap between complex data science models and daily retail operations. By providing instant visibility into customer segmentation and sales trajectories, the platform drives measurable efficiency gains, reduces inventory waste, and maximizes overall retail profitability.